MIL-F-16377/49A(SHIPS)
18 May 1973
SUPERSEDING
MIL-F-16377/49(SHIPS)
31 August 1971

MILITARY SPECIFICATION SHEET

FIXTURE, LIGHT, EXTENSION, FLUORESCENT, 8-WATT, 115-VOLT, 60 HERTZ, EXPLOSIONPROOF AND WATERTIGHT, DETAIL LIGHTING, SHIPBOARD USE

SYMBOL 306.2

The complete requirements for procuring the fixture described herein shall consist of this document and the latest issue of MIL-F-16377.



Figure 1 - Light, extension, fluorescent, symbol 306.2

(A) Denotes changes.

FSC 6230

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

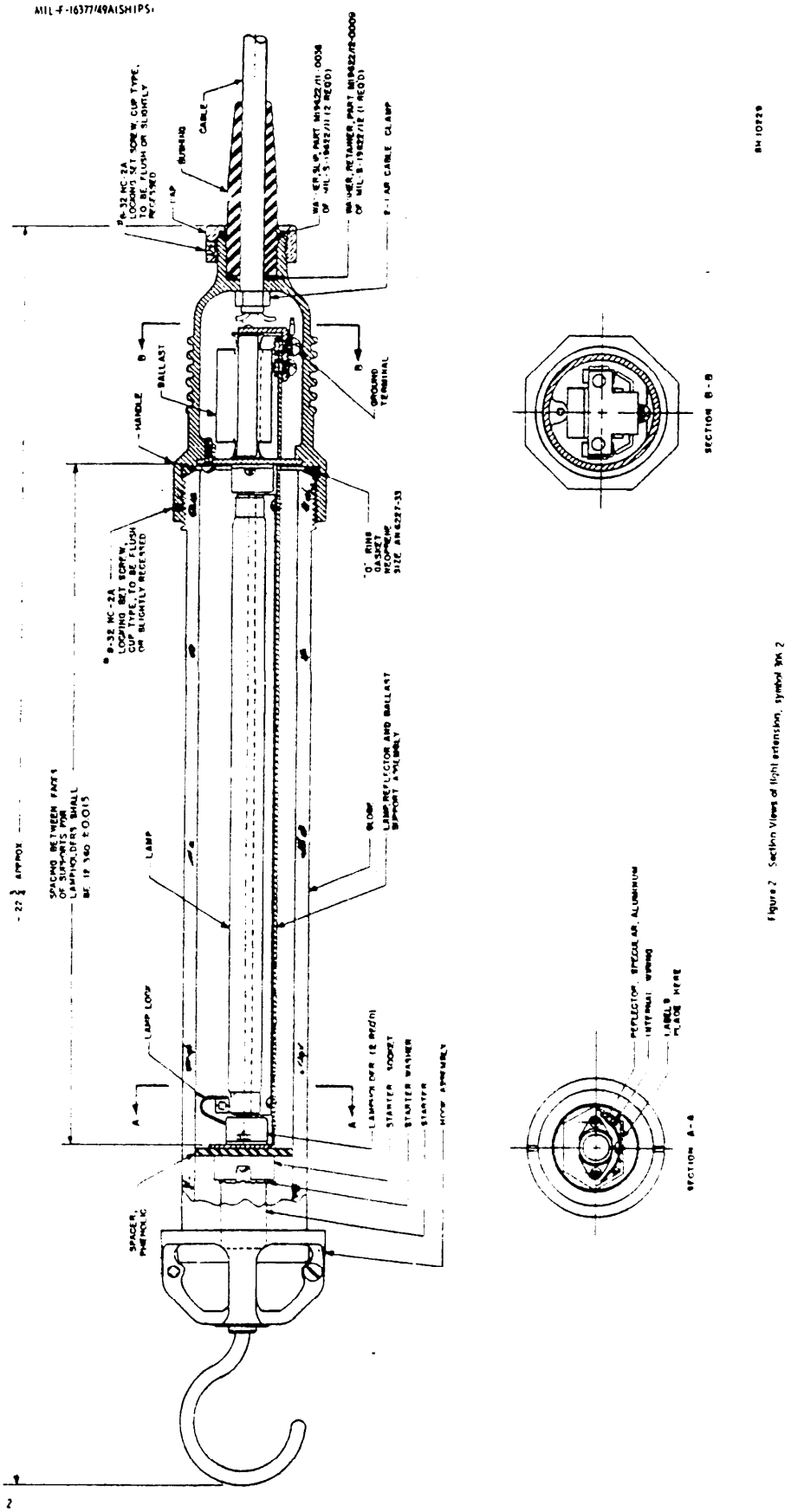


Figure 2 Section Views of light extension, symbol 3M 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

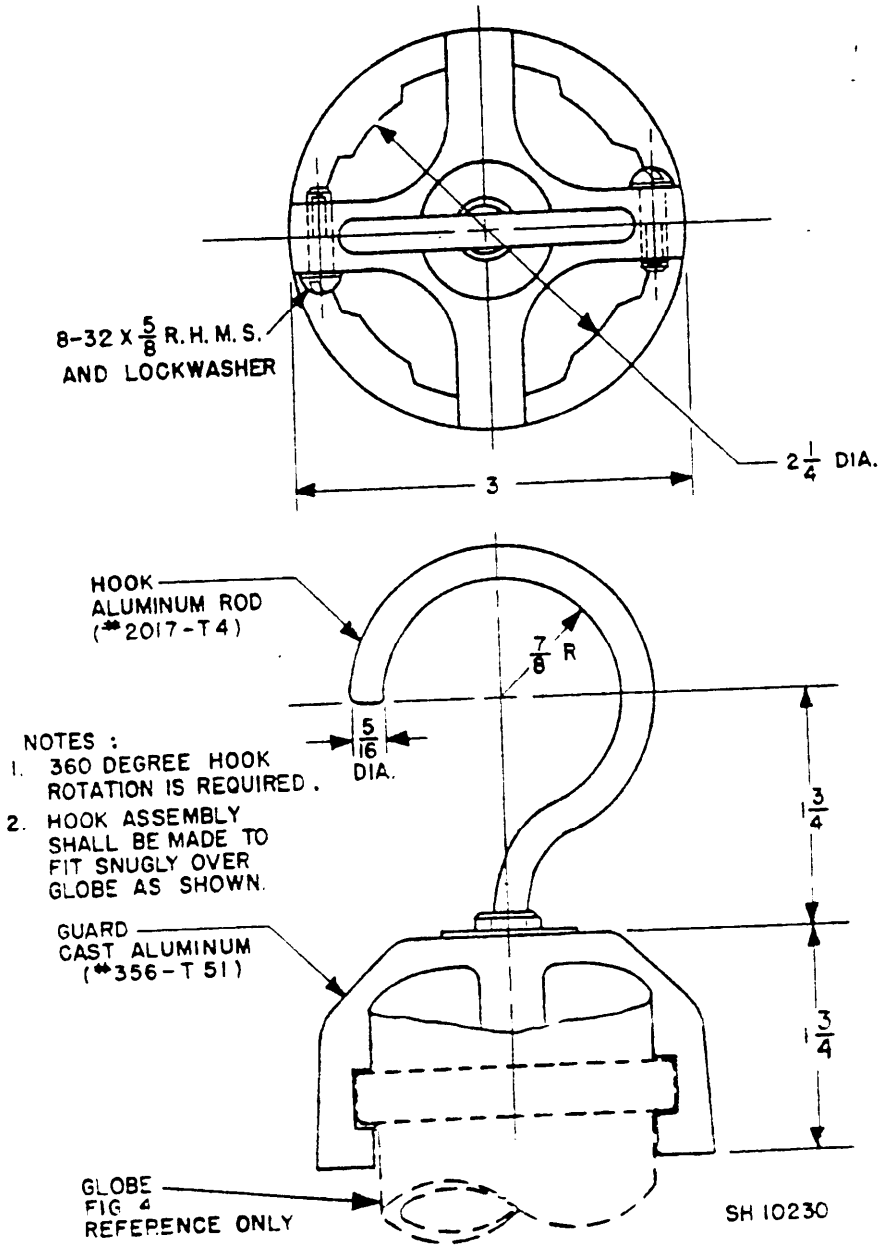


Figure 3 - Hook assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

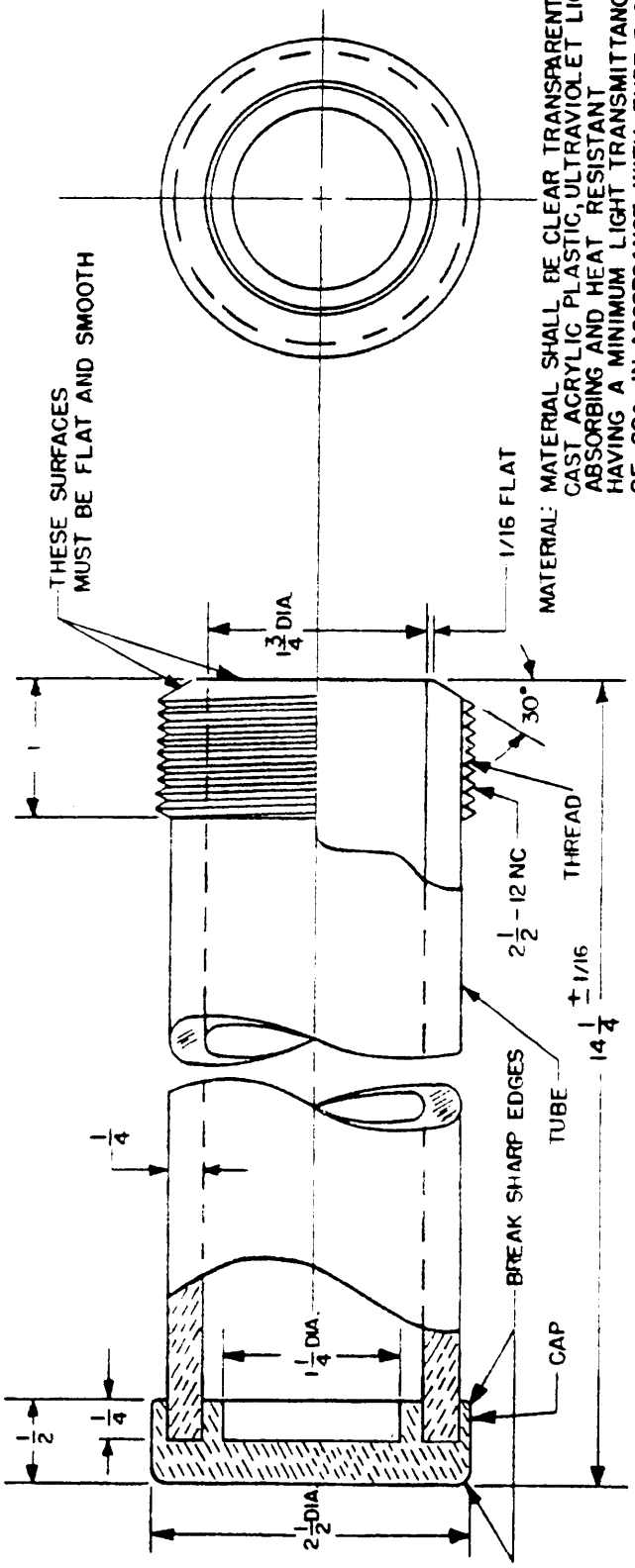


Figure 4 - Globe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

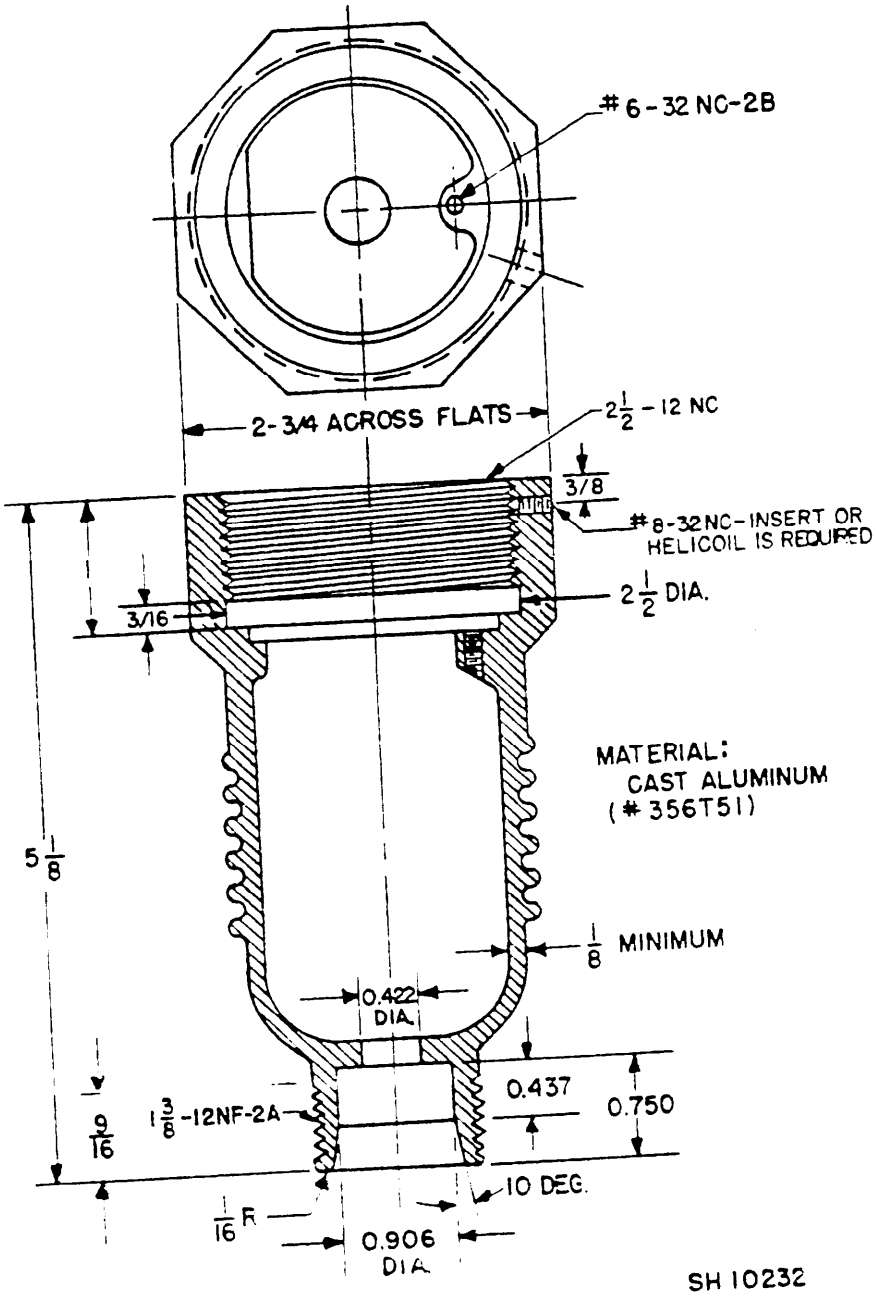
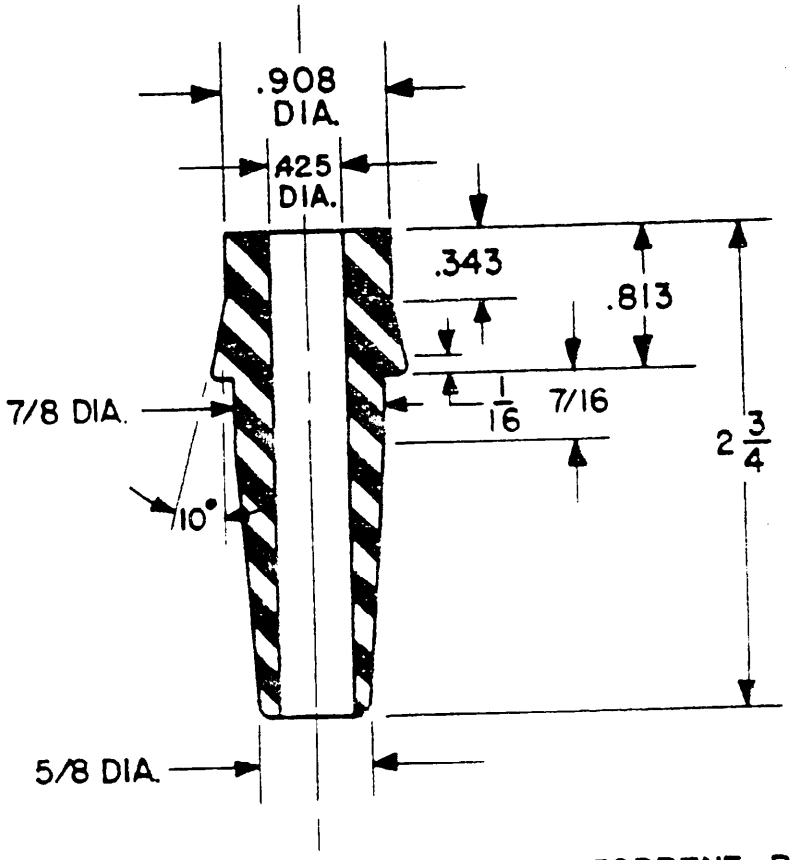


Figure 5 - Handle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.



MATERIAL: NEOPRENE RUBBER
35-45 DUROMETER

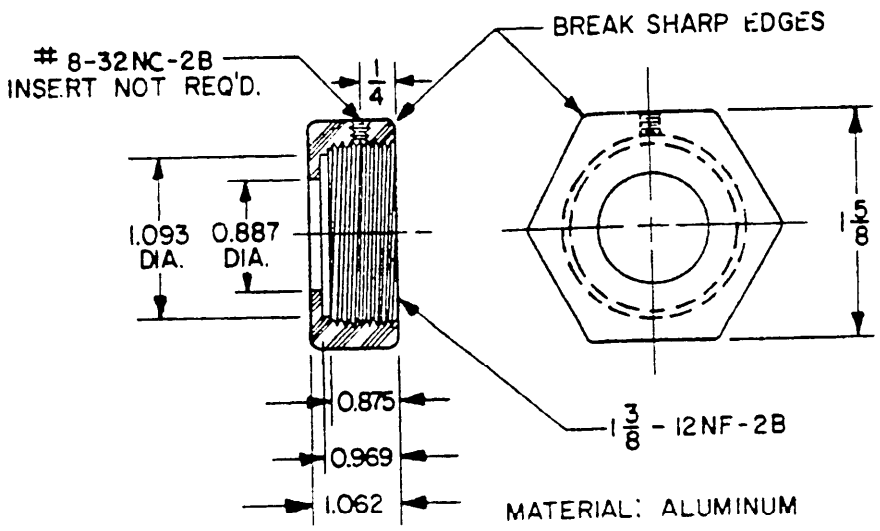
NOTE: DIMENSIONAL TOLERANCES
TO RMA CLASS 3

SH 10 233

Figure 6 - Bushing.

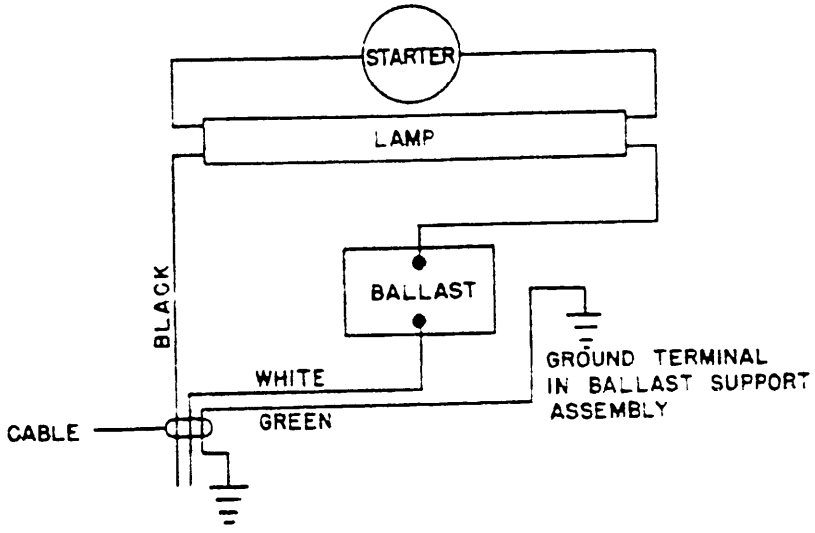
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

shall consist of the following assemblies:



SH 10234

Figure 7 - Cap.



SH 10235

Figure 8 - Wiring diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Type:	I
Class:	1
Symbol 306.2:	Light, extension, fluorescent, 8 watt, 115 volt, 60 Hertz, explosionproof and watertight.
Design:	
Electrical rating:	115 volts, 8-watt, 60 Hertz.
Dimensions and configurations:	See figures 1 through 8.
Metals:	All exposed metal parts shall be of nonsparking metal.
Enclosure:	The light shall be designed to be explosionproof and watertight and shall be so constructed that no parts will work loose in service. The enclosure shall be of substantial construction and shall be capable of withstanding the internal pressures resulting from explosions without bursting or loosening of the joints. The enclosure shall withstand without rupture or permanent distortion an internal hydrostatic test pressure of four times the maximum internal explosion pressure developed during the explosionproof test. The hydrostatic pressure shall be applied at a rate of about 100 pounds per square inch until required internal pressure is reached. The light shall be designed to be suitable for Class 1, Group D hazardous locations as defined by the National Electrical Code.
Cable:	The light shall be furnished with 50 feet of 3 conductor cable, type SO-16/3 or ST-16/3 in accordance with J-C-580. The third conductor shall be used for grounding.
Cable pull:	There shall be no strain at the cable connections or damage to the cable when a direct pull of 150 pounds is applied to the extension of the cable for 1 minute. The light shall be anchored by the hook.
Ambient temperature:	The light shall be capable of operating in an ambient temperature of 135°F. for 96 hours without damage or loosening of parts.
Rough usage:	The light shall be capable of withstanding the following rough usage test: The light shall be suspended in a vertical position with cable entrance up, by means of a short loop formed from a heavy solid wire. The cable shall be removed from the light and one end of the wire loop shall be connected to the light at its cable entrance. The other end of the wire loop shall be extended through a screweye secured to a solidly mounted vertical board, 2 inches thick. The free end of the light shall be pulled away from the board and then allowed to swing back against the board, or a wooden block secured thereto, for 9000 times at a rate of about 50 times per minute, the horizontal swing distance at the threaded engagement for the lamp replacement being about 6 inches. The enclosure and the cable entrance shall not be impaired nor shall threaded engagements be loosened during this test.
Globe:	Globe shall be as shown on figure 4. A dye penetrant test shall be made on all globes fabricated from three components to determine imperfections or cracks in welded joints. Globes shall be wiped clean after this test.
Drop:	The enclosure and the cable entrance shall not be impaired nor shall threaded engagements be loosened when the light is dropped to a concrete floor ten times from a height of 3 feet. For the first five drops during the test, the light, with cable connected as in service, shall be allowed to freely fall in a manner that would normally be the case if the device were to fall from a horizontal platform to the floor. In the remainder of the test, the light with cable shall be held at various angles about 3 feet above the concrete floor and then dropped to the floor.
Weight:	With cable - 10 pounds maximum. Without cable - 3 pounds maximum.
Labels:	The following labels shall be furnished in accordance with MIL-F-16377 and shall be attached as shown on figure 2: (a) Type I (one required). (b) Type IV. The information to be shown on this label shall be "USE THE FOLLOWING REPAIR PARTS ONLY". List all standard parts shown in table I by name and stock number only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (continued)

(c) Type V, except that the height of characters shall be not less than 1/16 inch. The information to be shown on this label shall be as follows:

"CAUTION

Disconnect from circuit before re-lamping. Keep all joints tight when in operation. If the connection to the electrical circuit is in an area where explosive gases are present, an explosionproof plug and matching receptacle shall be used. If the electrical connection is made outside of the hazardous area, a 3-conductor grounded plug may be used. The lamps shall not be replaced or the light disassembled in a hazardous location."

(A) (d) A warning label with red letters on white background with the following information:

"EXPLOSIONPROOF AND WATERTIGHT EXTENSION LIGHT FOR HAZARDOUS LOCATIONS CLASS 1, GROUP D. MAXIMUM OPERATING SURFACE TEMPERATURE: 50°C.

WARNING

FOR USE IN CLASS 1, GROUP D ATMOSHPERES ONLY IN WHICH AMONG OTHERS THE FOLLOWING ARE INCLUDED: GASOLINE, PETROLEUM NAPHTHA, ALCOHOL, ACETONE, LACQUER SOLVENT VAPORS AND NATURAL GAS.

CAUTION - MERCURY

USE APPROVED BY NAVSHIPS

Contains mercury in fluorescent lamp. Should the lamp break, DO NOT REPLACE until adequate overboard ventilation is established."

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

Table I - Standard parts.

Name	Specification or military part number	Federal stock number (FSN)	Remarks
Lamp	MS15588-3	6240-299-5546	Industry number F8T5/CW
(A) Globe	M16377/49-001	6230-451-6450	
Lamplock	M16377/40-001	6250-399-7039	
Lampholder	MIL-L-970/10	6250-377-8590	
Startersocket	MIL-L-970/13	6250-299-2738	
Starter (A)	BG, type III of W-S-755	6250-299-5962	Commercial Designation: FS5 Filling wax is not required
Ballast	M16377/45-001	6250-299-2735	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

Quality assurance: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:
Complete lights: See table II.
Globes: See table III.
Quality conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: See MIL-F-16377
Quality conformance tests: See MIL-F-16377

Table II - First article and comparison inspection for complete lights.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Rough usage	1/	1/	
Drop	1/	1/	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Ambient temperature	1/	1/	
Grounding circuit	3.6.13.3	4.8.11	
Leakage current	3.6.13.5	4.8.13	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Cable pull	1/	1/	
Magnetic permeability	3.4.8	4.8.18	
Salt spray	3.3.15	4.8.10	
Enclosure effectiveness	3.5.11	4.8.15	Explosion-proof
Hydrostatic pressure	1/	1/	
Dye penetrant test for globes	1/	1/	

1/ As specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

Table III - First article and comparison inspection for globes.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Dye penetrant	1/	1/	

1/ As specified herein.

General Information: (a) Assignment of Military part numbers and general information is shown in table IV.

Table IV - Part number and general information.

Military part number M16377/49	Item name	Federal Stock Number (FSN)
-306.2	Complete light, symbol 306.2	6230-244-3996
-001	Globe	6230-451-6450
-002	Hook assembly	
-003	Handle	
-004	Bushing	

(b) This light is intended to be used as a general purpose extension light for detail illumination onboard ship and as a work light in areas where explosive gases may be present.

Preparing activity:
Navy - SH
(Project 6230-N196)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SHIP ENGINEERING CENTER (CODE 6124)
CENTER BUILDING
PRINCE GEORGE'S CENTER
HYATTSVILLE, MD 20782

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS POINT, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.